ALGORITHM FOR THE PROJECT

1. **Start** the program.
2. **Display** the application title: "LED RESISTOR VALUE CALCULATOR SYSTEM".
3. **Prompt and Input** the total number of LED circuits to be calculated (numberOfCircuits)
4. **Initialize a loop** to execute from 1 up to numberOfCircuits.
5. Within the loop, **Prompt and Input**: ledName, supplyVoltage, forwardVoltage, and ledCurrentMilliAmps
6. **Perform Data Validation (Decision Check):**

Check if supplyVoltage ≤ forwardVoltage OR ledCurrentMilliAmps ≤ 0.

- Branch 6a (If YES/Invalid parameters): Set designStatus = "Invalid Design". Skip all math operations and safety checks, and proceed straight to Step 10.
- **Branch 6b (If NO/Valid parameters):** Execute the technical processing calculations sequentially:
- • **Step 6b.1:** Convert LED current from milliamperes to amperes: ledCurrentAmps = ledCurrentMilliAmps / 1000.
- • **Step 6b.2:** Calculate the resistor value in ohms: resistorValue = (supplyVoltage - forwardVoltage) / ledCurrentAmps.
- • **Step 6b.3:** Calculate the minimum resistor power rating in watts: resistorPower = ledCurrentAmps * ledCurrentAmps * resistorValue.

7. **Evaluate Safety Threshold (Decision Check):** Check if the desired LED current is greater than 30 Ma

- **Step 8a (If YES):** Set designStatus = "Warning: LED current may be too high for a standard LED.".

- **Step 8b (If NO):** Set designStatus = "Design Accepted".

9. **Output and Display** the calculated parameters (resistorValue, resistorPower) and the final designStatus directly to the terminal console.

10. **Loop Condition Evaluation (Decision Check):** Check if all specified circuits have been processed.

    - **If NO (More circuits remaining):** Route the execution path back to the loop initialization point (**Step 3**).

    - **If YES (All iterations completed):** Exit the loop and proceed downward.

11. **Save Report:** Export and write the completed report data for all processed circuits into an external text file named led_resistor_report.txt

12. **End** the programme.